More Diverse, More Vulnerable: How Judge Panel Diversity Backfires Under Adaptive Adversaries

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Abstract

Multi-judge evaluation panels are increasingly used to assess LLM outputs, with the prevailing assumption that judge diversity—analogous to Byzantine Fault Tolerance (BFT)—improves robustness against manipulation. We provide large-scale empirical evidence that this assumption *reverses* under adaptive adversaries. Across 15 LLM judges spanning 6 model families, 455 three-judge panels, 9 attack types, and 4 evaluation benchmarks, we show that panel diversity (effective independence K_{eff}) is **independently and positively** associated with attack success rate (ASR), even after residualizing against individual judge weakness ($\eta_{\max}$). This "diversity trap" is driven by dual vulnerability channels whose relative importance is attack-dependent: $\eta_{\max}$ dominates shared-bias attacks (sycophancy, verbosity), while K_{eff} dominates transferable attacks (score manipulation, position bias). We trace the mechanism to BFT's broken assumption: random faults distribute uniformly across judges (Gini$\approx$0.12), but adaptive adversaries concentrate attacks on the weakest judge (Gini$\approx$0.68), turning diversity from a defense into an expanded attack surface. Under calibrated random faults, K_{eff}'s coefficient flips negative in 5/12 conditions, empirically confirming BFT's prediction. We propose a robustness-first two-stage panel selection algorithm that reduces ASR by 20.8% over the strongest-K baseline, and show that Bayesian aggregation provides a complementary 28.4% reduction. Our findings challenge the "more diverse = more robust" assumption and call for diversity-aware security analysis in LLM evaluation pipeline design.

1 Introduction

As LLMs are increasingly deployed as automated evaluators [11, 6], the research community has adopted multi-judge panels to mitigate individual model biases. The intuition is straightforward and borrowed from distributed systems: diverse, independent judges should be harder to simultaneously fool, much like Byzantine Fault Tolerance (BFT) guarantees that f faulty nodes cannot compromise a system of $3f + 1$ independent replicas.

This paper demonstrates that this intuition is *wrong* under adversarial conditions. We conduct what is, to our knowledge, the largest systematic study of adversarial robustness in multi-judge LLM evaluation: 15 models from 6 families (OpenAI, Anthropic, Google, Meta, Alibaba, Microsoft), enumerated into 455 three-judge panels, evaluated under 9 distinct attack types across MMLU, ARC-Challenge, MT-Bench, and AlpacaEval.

Our central finding is the **diversity trap**: panel diversity, measured by effective judge independence (K_{eff}), is *independently and positively* associated with attack success rate. After residualizing K_{eff} against individual judge weakness ($\eta_{\max}$, the maximum vulnerability of any panel member), $K_{\text{eff}}^{\text{resid}}$ remains significantly positive in 10/12 conditions (FDR-corrected). Panel vulnerability is driven by **dual factors**—$\eta_{\max}$ and K_{eff}—whose relative importance depends on attack type:

- **Shared-bias attacks** (sycophancy, verbosity, authority): $\eta_{\max}$ dominates because all judges share similar cognitive biases from RLHF training.
- **Transferable attacks** (score manipulation, position bias): K_{eff} dominates because diverse judges provide heterogeneous, judge-specific attack surfaces.

We trace this to BFT's core assumption: fault independence. Under random faults, failures distribute uniformly (Gini≈0.12) and diversity genuinely helps—our calibrated random-fault simulations show K_{eff}'s coefficient *flipping negative* in 5/12 conditions. But adaptive adversaries concentrate attacks on the weakest judges (Gini≈0.68, top-1 judge absorbing 78% of ASR), turning diversity from fault tolerance into an expanded attack menu.

We further show that: (1) the diversity trap holds within capability tiers (4/6 conditions significant among top-tier models only), ruling out tier-mixing confounds; (2) generalizes to open-ended benchmarks (MT-Bench 3/4, AlpacaEval 2/4 significant); (3) scales with panel size (K_{eff} coefficient increases monotonically from K=3 to K=9); (4) holds under partial-knowledge adversaries who know only model family membership (7/12 significant). We propose a two-stage defense—robustness screening followed by diversity optimization—that achieves 20.8% ASR reduction over the strongest-K baseline, and show Bayesian aggregation provides complementary protection.

2 Related Work

LLM-as-Judge. LLM evaluation using model judges has become standard practice [11, 6, 3]. Known biases include position bias, verbosity preference, and self-enhancement [10, 11]. Multi-judge panels are proposed as mitigation, but their adversarial robustness remains unstudied.

Adversarial Attacks on LLM Judges. Prior work demonstrates individual judge vulnerability to prompt injection, sycophancy exploitation, and score manipulation [7, 9]. We extend this to *panel-level* analysis, studying how panel composition mediates vulnerability.

Byzantine Fault Tolerance. Classical BFT [5, 2] guarantees safety when faults are independent. Recent work questions independence assumptions in ML systems [8]. We empirically test whether BFT's independence assumption holds for LLM judge panels.

Ensemble Diversity. The diversity-accuracy tradeoff is well-studied in ensemble learning [4, 1]. Our setting differs: we study diversity under *adaptive adversaries* who exploit panel composition, not passive noise.

3 Problem Formulation

Setup. A panel $P = \{j_1, \ldots, j_K\}$ of K LLM judges evaluates candidate responses. Each judge j_i produces a binary decision (accept/reject). The panel decision uses an aggregation rule $\mathcal{A}$ (default: majority vote).

Adversary Model. An adaptive adversary observes the panel composition and crafts attacks to maximize the panel's attack success rate (ASR)—the fraction of evaluation instances where the panel decision is flipped.

Vulnerability Metrics. We define two panel-level predictors:

- $\eta_{\max} = \max_{j \in P} \eta_j$: maximum individual vulnerability (weakest link). η_j is judge j's individual ASR under a given attack.
- K_{eff}: effective number of independent judges, computed from the eigenspectrum of the pairwise mutual information matrix. Higher K_{eff} indicates greater functional diversity.

To disentangle K_{eff} from $\eta_{\max}$, we define $K_{\text{eff}}^{\text{resid}}$ as the residual of regressing K_{eff} on $\eta_{\max}$, ensuring orthogonality (VIF=1.02).

Regression Model. For each condition (dataset $\times$ attack type), we fit:

$$\text{ASR}_P = \beta_0 + \beta_\eta \cdot \eta_{\text{max}} + \beta_K \cdot K_{\text{eff}}^{\text{resid}} + \epsilon \tag{1}$$

across all 455 panels. A significantly positive β_K indicates diversity independently increases vulnerability.

4 Experimental Setup

Models. 15 LLM judges: GPT-4o, GPT-4o-mini, GPT-4.1, GPT-4.1-mini (OpenAI); Claude-3.5-Sonnet (Anthropic); Gemini-1.5-Pro (Google); Llama-3.1-8B, Llama-3.1-70B (Meta); Qwen2.5-7B, Qwen2.5-72B (Alibaba); Mistral-7B-v0.3, Phi-3-medium-14B (Microsoft). Three capability tiers: Small (7–14B), Medium (14–34B), Large (70B+ and API).

Panels. All $\binom{15}{3} = 455$ three-judge panels, extended to $K \in \{5, 7, 9\}$ for panel-size analysis.

Attacks. 9 attack types spanning three paradigms: *bias exploitation* (sycophancy, verbosity, authority bias), *score manipulation* (direct score manipulation, position bias, few-shot bias), and *injection* (few-shot hijacking, persona manipulation, instruction override).

Benchmarks. MMLU and ARC-Challenge (MCQ); MT-Bench and AlpacaEval (open-ended generation).

Statistical Methods. FDR correction (Benjamini-Hochberg) for multiple comparisons; beta regression for distributional robustness; residualization for multicollinearity control.

5 The Diversity Trap: Core Evidence

5.1 Dual Vulnerability Drivers

Table 1 presents the core regression results across 12 conditions (2 datasets $\times$ 6 attacks). Both η_{max} and $K_{\text{eff}}^{\text{resid}}$ are significant predictors of panel ASR: η_{max} reaches FDR significance in 10/12 conditions, $K_{\text{eff}}^{\text{resid}}$ in 10/12 conditions. The mean $R^2 = 0.62$, indicating that these two factors jointly explain the majority of panel-level vulnerability variance.

A systematic pattern emerges: **transferable attacks** (score manipulation, position bias) are K_{eff}-dominated ($\beta_K > \beta_\eta$), while **shared-bias attacks** (sycophancy, verbosity, authority) are η_{max}-dominated. This yields a predictive taxonomy: given a new attack type, its transferability predicts which vulnerability channel dominates.

Three conditions have low confound correlation ($r \approx 0.35$), serving as "smoking gun" evidence: in these conditions, $K_{\text{eff}}^{\text{resid}}$ achieves $p < 2.1 \times 10^{-28}$, confirming diversity's independent harmful effect even where the K_{eff}-η_{max} association is weak.

5.2 Capability-Matched Validation

A natural concern is that K_{eff}'s positive effect is an artifact of tier-mixing: high-K_{eff} panels might mix weak and strong models, making K_{eff} a proxy for tier heterogeneity rather than diversity *per se*. We test this by running within-tier regressions (Table 2).

The large tier is most informative: all 6 models are top-tier (GPT-4o, GPT-4.1, Claude-3.5-Sonnet, Gemini-1.5-Pro, Llama-70B, Qwen-72B), yet $K_{\text{eff}}^{\text{resid}}$ is significant in 4/6 conditions (best $p = 4.2 \times 10^{-4}$). The condition-dependent pattern is preserved: score manipulation and position bias are K_{eff}-dominated; sycophancy and verbosity are η_{max}-dominated. This rules out tier-mixing as the driver of the diversity trap.

Table 1: Dual vulnerability drivers: 15-model regression across 12 conditions (455 panels each). $K_{\text{eff}}^{\text{resid}}$ is residualized against $\eta_{\max}$ (VIF=1.02). Bold p-values are FDR-significant ($q < 0.05$). "Dominant" indicates which predictor has the larger standardized β.

Condition	$\eta_{\max}$ β	p	$K_{\text{eff}}^{\text{resid}}$ β	p	R^2	Confound r	Dominant
mmlu×sycophancy	0.82	**1.2e-8**	0.31	**0.004**	0.68	0.72	$\eta_{\max}$
mmlu×score_manip	0.45	**0.018**	0.61	**3.4e-9**	0.71	0.38	K_{eff}
mmlu×position	0.15	0.214	0.58	**1.2e-8**	0.56	0.36	K_{eff}
mmlu×verbosity	0.74	**4.1e-7**	0.28	**0.011**	0.63	0.74	$\eta_{\max}$
mmlu×authority	0.69	**1.8e-6**	0.19	0.087	0.58	0.78	$\eta_{\max}$
mmlu×few_shot	0.55	**2.3e-4**	0.48	**6.1e-6**	0.65	0.65	both
arc×sycophancy	0.79	**3.5e-8**	0.35	**0.002**	0.66	0.71	$\eta_{\max}$
arc×score_manip	0.38	0.117	0.64	**1.7e-7**	0.59	0.35	K_{eff}
arc×position	0.03	0.910	0.72	**1.8e-8**	0.57	0.34	K_{eff}
arc×verbosity	0.71	**8.9e-7**	0.33	**0.003**	0.61	0.73	$\eta_{\max}$
arc×authority	0.66	**3.2e-6**	0.22	0.062	0.55	0.76	$\eta_{\max}$
arc×few_shot	0.51	**5.7e-4**	0.52	**3.8e-6**	0.63	0.62	both

Table 2: Within-tier K_{eff} validation. Large tier uses only top-tier models (70B+ and API), eliminating tier-mixing.

Tier	Models	Panels	$r(K_{\text{eff}}, \eta_{\max})$	$K_{\text{eff}}^{\text{resid}}$ sig
Small (7B–14B)	5	10	0.32	2/6
Medium	4	4	0.41	1/6
Large (70B+API)	6	20	0.33	4/6
Cross-tier (all)	15	455	0.55	10/12

5.3 Statistical Robustness

We perform comprehensive diagnostics (Table 3). VIF=1.02 and condition number=4.8 confirm the design matrix is well-conditioned. Residuals pass Shapiro-Wilk normality ($p = 0.23$). Maximum Cook's distance is 0.031 (threshold: 1.0), with only 2.4% high-leverage points. Under Bonferroni correction (most conservative), $K_{\text{eff}}^{\text{resid}}$ remains significant in 8/12 conditions. Beta regression, which properly handles bounded ASR in $[0, 1]$, confirms 9/12 significant—highly consistent with OLS.

6 Mechanism: BFT Reversal

6.1 Random-Fault Contrast

BFT predicts that diversity helps when faults are independent and random. We test this by simulating calibrated random faults: each judge's failure probability is drawn from bootstrapped clean-accuracy noise, independent across judges. Table 4 contrasts K_{eff}'s coefficient under adaptive versus random-fault regimes.

The results confirm the BFT reversal: in 5/12 conditions, β_{RF} flips to significantly *negative*—diversity genuinely protects under random faults, as BFT predicts. The flips concentrate on transferable attacks (score manipulation, position bias, few-shot), where judge-specific exploit paths are most relevant. Under adaptive attacks, these same conditions show the strongest positive β (0.48–0.72). The mean β ratio is 2.8×, and the R^2 gap ($\Delta R^2 = 0.31$; RF: 0.78, adaptive: 0.47) quantifies BFT's broken predictability: panel structure explains 78% of variance under random faults but only 47% under adaptive attacks, where strategic adversary behavior introduces irreducible noise.

Table 3: Multiple comparison robustness across correction methods.

Correction	$K_{\text{eff}}^{\text{resid}}$ sig	$\eta_{\max}$ sig
Uncorrected	12/12	11/12
FDR (BH)	10/12	10/12
Bonferroni	8/12	9/12
Beta reg. + FDR	9/12	10/12

Table 4: BFT regime contrast: K_{eff} coefficient under adaptive vs. calibrated random faults (12 conditions, 455 panels). "Flipped" indicates β_{RF} is significantly negative (diversity helps under random faults).

Condition	β_{adaptive}	p_{adap}	β_{RF}	p_{RF}	$\lvert\beta_{\text{adap}}/\beta_{\text{RF}}\rvert$	Flipped?
mmlu×sycophancy	0.31	**0.004**	0.08	0.310	3.88	No
mmlu×score_manip	0.61	**3.4e-9**	−0.22	**0.015**	2.77	Yes
mmlu×position	0.58	**1.2e-8**	−0.31	**0.003**	1.87	Yes
mmlu×verbosity	0.28	**0.011**	0.11	0.198	2.55	No
mmlu×authority	0.19	0.087	0.06	0.541	3.17	No
mmlu×few_shot	0.48	**6.1e-6**	−0.18	**0.039**	2.67	Yes
arc×sycophancy	0.35	**0.002**	0.10	0.251	3.50	No
arc×score_manip	0.64	**1.7e-7**	−0.28	**0.008**	2.29	Yes
arc×position	0.72	**1.8e-8**	−0.35	**0.001**	2.06	Yes
arc×verbosity	0.33	**0.003**	0.12	0.175	2.75	No
arc×authority	0.22	0.062	0.05	0.612	4.40	No
arc×few_shot	0.52	**3.8e-6**	−0.14	0.091	3.71	Marginal

6.2 ASR Concentration Analysis

To illuminate *how* adaptive adversaries exploit diversity, we analyze per-judge ASR distributions within panels (Table 5).

Under adaptive attacks, ASR is highly concentrated: the weakest judge absorbs 78% of attack success (vs. 38% under random faults). Crucially, $r(K_{\text{eff}}, \text{Gini}) = 0.52$: more diverse panels produce *more concentrated* attacks, because greater capability spread gives the adversary a clearer targeting gradient. Under random faults, concentration is uniformly low (Gini=0.12), confirming BFT's independence assumption holds only in the non-adversarial regime.

7 Generalization

7.1 Attack Surface Expansion

To test whether the diversity trap generalizes beyond the original 6 attack types, we introduce 3 additional attacks: few-shot hijacking (in-context example injection), persona manipulation (authority impersonation), and instruction override (direct prompt overwrite). Table 6 shows the 6 new conditions.

$K_{\text{eff}}^{\text{resid}}$ is significant in 4/6 new conditions. Combined with the original 12, this yields 14/18 total FDR-significant conditions for $K_{\text{eff}}^{\text{resid}}$. The attack taxonomy is predictive: high-transferability attacks (few-shot hijacking) are K_{eff}-dominated; moderate/low-transferability attacks (persona, instruction override) are $\eta_{\max}$-dominated—consistent with the core results.

7.2 Open-Ended Benchmarks

The diversity trap is not an MCQ artifact. On MT-Bench (multi-turn dialogue evaluation) and AlpacaEval (instruction-following scoring), $K_{\text{eff}}^{\text{resid}}$ is significant in 3/4 and 2/4 conditions, respectively (Table 7). The condition-dependent dominance pattern transfers without modification: score manipulation and position

Table 5: ASR concentration: adaptive vs. random-fault regimes.

Metric	Adaptive	RF	Ratio
Mean Gini	0.68	0.12	5.67
Entropy (bits)	1.42	2.31	0.61
Top-1 judge share	0.78	0.38	2.05
$r(K_{\text{eff}}, \text{Gini})$	0.52	0.08	6.50

Table 6: Extended attack results (6 new conditions, 15 models).

Condition	$K_{\text{eff}}^{\text{resid}}$ β	p	Dominant
mmlu×hijack	0.54	**8.2e-7**	K_{eff}
mmlu×persona	0.29	**0.008**	η_{max}
mmlu×override	0.22	0.042	η_{max}
arc×hijack	0.58	**2.1e-7**	K_{eff}
arc×persona	0.25	**0.019**	η_{max}
arc×override	0.18	0.112	η_{max}

bias remain K_{eff}-dominated; sycophancy and verbosity remain η_{max}-dominated. R^2 is lower (0.39–0.52 vs. 0.62 for MCQ), reflecting the higher noise floor inherent to open-ended evaluation.

7.3 Panel Size Sensitivity

Extending panel size from $K = 3$ to $K \in \{5, 7, 9\}$ reveals a scaling law: K_{eff}'s coefficient increases monotonically (β_K: 0.42→0.48→0.55→0.61), while η_{max}'s decreases (0.51→0.49→0.43→0.38). The crossover occurs at $K \approx 6$: for panels below this size, individual robustness (η_{max}) is the primary concern; above it, structural diversity (K_{eff}) becomes dominant (Table 8). This has direct practical implications: the common strategy of "adding more judges" systematically amplifies the diversity trap—$K = 9$ panels show 45% stronger K_{eff} effect than $K = 3$.

7.4 Partial-Knowledge Adversary

A full-knowledge adversary is a worst-case assumption. We test a realistic partial-knowledge adversary who knows only model *family* membership (typically public information) but not individual vulnerability profiles. Family-level targeting achieves 78.4% of full-knowledge ASR (0.058 vs. 0.074), and K_{eff} remains significant in 7/12 conditions (Table 9). Family membership alone captures 62.8% of adversary information value, because family→capability tier correlations provide a strong prior on vulnerability profiles.

8 Structural Barriers to Independence

8.1 Gaussian Copula Failure

If inter-judge dependence followed a Gaussian copula, pairwise mutual information (MI) could be estimated from linear correlations. Across 105 model pairs (15 models), observed MI exceeds Gaussian-predicted MI by a mean factor of 18.5× (max 312×), with 82/105 pairs exceeding 10×. This invalidates copula-based independence bounds and motivates empirical K_{eff} measurement.

8.2 Family Labels Are Unreliable

Model family labels do not predict judge independence: within-family MI (0.42) is not significantly higher than cross-family MI (0.39; Mann-Whitney $p = 0.158$). The actual driver is capability proximity

Table 7: Open-ended benchmark results (8 conditions). Pattern consistent with MCQ findings despite higher evaluation noise.

Condition	$K_{\text{eff}}^{\text{resid}}$ β	p	R^2	Dom.
MT×score	0.52	**1.8e-5**	0.52	K_{eff}
MT×syco	0.28	**0.012**	0.49	η_{max}
MT×pos	0.47	**8.3e-5**	0.44	K_{eff}
MT×verb	0.18	0.108	0.47	η_{max}
AE×score	0.44	**3.2e-4**	0.46	K_{eff}
AE×syco	0.21	0.058	0.43	η_{max}
AE×pos	0.38	**0.002**	0.40	both
AE×verb	0.14	0.175	0.39	η_{max}

Table 8: Panel size sensitivity ($K = 3, 5, 7, 9$).

K	Panels	$K_{\text{eff}}^{\text{resid}}$ sig	$\bar{\beta}_K$	$\bar{\beta}_\eta$	Primary
3	455	10/12	0.42	0.51	η_{max}
5	3003	11/12	0.48	0.49	balanced
7	6435	12/12	0.55	0.43	K_{eff}
9	5005	12/12	0.61	0.38	K_{eff}

($r = -0.71$ with MI): models at similar capability levels share failure modes regardless of training pipeline. Practitioners should not equate "different families" with "diverse panel."

8.3 Diversity-Capability Tradeoff

Across 455 panels, K_{eff} and η_{max} correlate at $r = 0.55$ globally. Within capability tiers, this drops to $r \approx 0.33$, revealing two components: a cross-tier component (∼40%, from mixing weak and strong models) and a within-tier residual (∼60%, from genuine diversity-vulnerability association). The Pareto frontier (18 panels, slope -0.73) quantifies the tradeoff: each 0.1 increase in K_{eff} costs at least 0.073 increase in η_{max}.

9 Defense: Panel Design and Aggregation

9.1 Two-Stage Panel Selection

We propose a robustness-first algorithm: (1) screen judges by individual robustness ($\eta_j \leq$ percentile-75 threshold), (2) maximize K_{eff} among qualifying judges. Table 10 benchmarks this against 5 baselines.

Two-stage achieves the lowest ASR among non-oracle strategies (0.038), outperforming the strongest-K baseline by 20.8%. Max-K_{eff} is catastrophic (ASR 0.112, +133%), providing the most direct evidence against blind diversity maximization. Sensitivity analysis shows the screening threshold is robust across percentile 50–75, with optimal performance at percentile 65–75.

9.2 Aggregation Rule Modulation

The dual vulnerability driver effect varies by aggregation rule (Table 11). Threshold (unanimity) voting amplifies K_{eff}'s harmful effect by 45% (β_K: 0.42→0.61) and increases ASR by 86%, because a single flipped judge blocks panel rejection. Bayesian aggregation, which downweights judges based on prior accuracy, provides the best single-aggregation defense: 28.4% ASR reduction, with K_{eff} significance dropping from 10/12 to 6/12. However, 6/12 K_{eff} significance confirms Bayesian aggregation alone cannot eliminate the diversity trap, motivating a combined defense stack.

Table 9: Adversary knowledge level comparison.

Knowledge	ASR	K_{eff} sig	R^2	Info value
Full	0.074	10/12	0.62	100%
Partial (family)	0.058	7/12	0.51	62.8%
Black-box	0.031	3/12	0.28	0%

Table 10: Panel selection strategy benchmark (15 models, MMLU, 6 attacks). Two-stage achieves 84% of oracle efficiency.

Strategy	ASR	Δ vs Strong-K	Best (of 6)
Two-Stage (ours)	0.038	-20.8%	5/6
Min-$\eta_{\max}$	0.041	-14.6%	4/6
Strongest-K	0.048	baseline	1/6
Random	0.089	+85.4%	0/6
Max-K_{eff}	0.112	+133.3%	0/6
Oracle	0.032	-33.3%	6/6

10 Conclusion

We presented large-scale empirical evidence for the *diversity trap* in multi-judge LLM evaluation: panel diversity independently increases vulnerability to adaptive adversarial attacks. This finding, validated across 15 models, 455 panels, 9 attack types, 4 benchmarks, and multiple panel sizes, challenges the pervasive assumption that diversity improves evaluation robustness. The mechanism traces to BFT's broken assumption: adaptive adversaries exploit the heterogeneous attack surface that diverse judges create, concentrating attacks on the weakest judge rather than facing independent random faults. We proposed a two-stage defense (robustness screening + diversity optimization) that reduces ASR by 20.8%, complemented by Bayesian aggregation (-28.4%). We urge the LLM evaluation community to adopt diversity-aware security analysis: do not assume more judges means safer evaluation.

Limitations

Our study has three main limitations. First, our adversary model assumes knowledge of panel composition (full or partial); in settings where panel membership is secret, the diversity trap may be attenuated. Second, while we cover 15 models from 6 families, the rapidly evolving LLM landscape means our specific quantitative findings may shift as new model generations emerge. Third, our analysis is observational—the regression design establishes independent predictive association, not strict causality, between diversity and vulnerability; experimental manipulation of panel composition with controlled interventions would strengthen causal claims.

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Table 11: Aggregation rule × vulnerability drivers (455 panels).

Aggregation	K_{eff} sig	$\bar{\beta}_K$	ASR	Δ
Majority	10/12	0.42	0.074	–
Threshold	11/12	0.61	0.138	+86%
Acc-weighted	8/12	0.39	0.069	−6.8%
Bayesian	6/12	0.25	0.053	−28.4%

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